

M23BEC703

Transmitting / Receiving voice and data using electromagnetic waves in free space is called as Wireless communication.

- James Clerk Maxwell is the father of WC.
- Experimented by Marconi [invented Radio (wireless)]
- J.C. Bose also experimented but failed to publish.

* Light is an electromagnetic energy.

Types of Wireless communication.

1. Mobile WC
2. Portable WC - Ex: WiFi, Bluetooth
3. fixed WC - Fixed wireless landline.

Challenges in developing WC:

- Build or design efficient hardware (low power Tx (Rx))
- Integrated services.

Note: Propagation environment itself is called as a channel.

Network: Base station, user equipment / Mobile station, Base station controller, Mobile switching centre, PSTN (Public Switch Telephone Network).

Wireless generations: analog cellular system.

1. First generation: analog

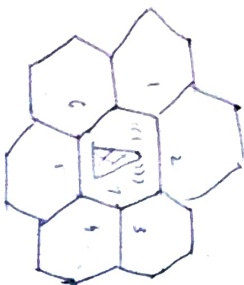
- started in 1980s
- Provided mobility.
- Analog in nature (No security)
- Multi users can access
- large cells omni directional antennas
- speech transmitted via FM and controlled through FSK
- seven cell pattern

Why analog wasn't enough

- capacity
- security
- voice quality
- no data services

2. Second generation: Digital cellular system

- gave digital efficiency & data
- digital in nature
- 2 paths — $\begin{cases} \rightarrow \text{TDMA (GSM)} \\ \rightarrow \text{CDMA} \end{cases}$



- TDMA (Time division Multiple Access)
- division of frequency w/ time, No simultaneous access
- CDMA (code division Multiple Access)
- Users can access data simultaneously using unique codes (interference is zero)

Therefore SNR increases (signal interference noise ratio)

Note: $y + H_2 + n$

y → Received signal

H → channel

n → noise

The four major 2G standards:

standard	Forward (MHz)	Reverse (MHz)	Access	Region
IS-54/115-136	869-960	824-849	TDMA/FDD	US
GSM	935-960	890-915	TDMA/FDD	Europe/USA
PDC	910-956	810-826	TDMA/FDD	Japan
IS-95	819-894	824-849	CDMA/FDD	US/Asia

Three benefits of 2G over 1G

- Digitally encrypted calls.
- Far greater spectrum efficiency → more subscribers
- spectral efficiency per user

GSM: 25 KHz/user

IS-54: 10 KHz/user

TDG: 8.33 KHz/user

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3. 2.5G (GPRS)

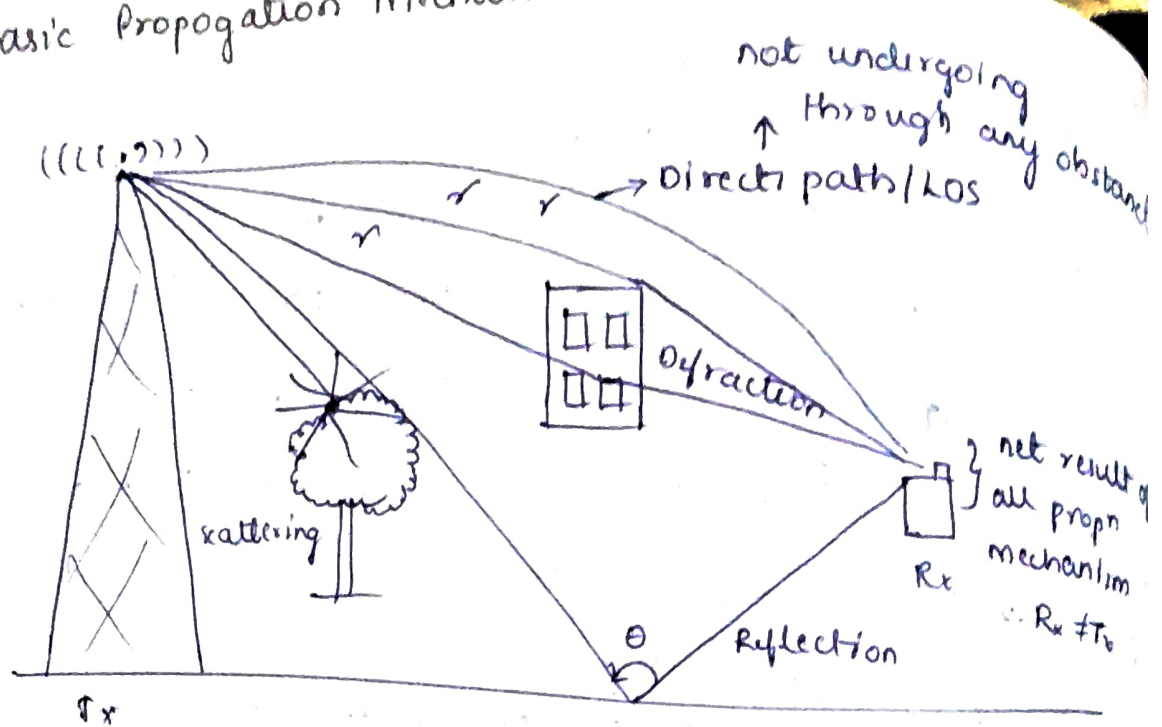
- voice transmitted through packet
- EDGE - Enhanced data rate for GSM evolution

Note: Rx & Tx

4. 3G

- gave broadband multimedia (shared for certain amount of people)
- wideband CDMA (huge amount of band shared for all)
- integrating multimedia
- Video calling - 2010's

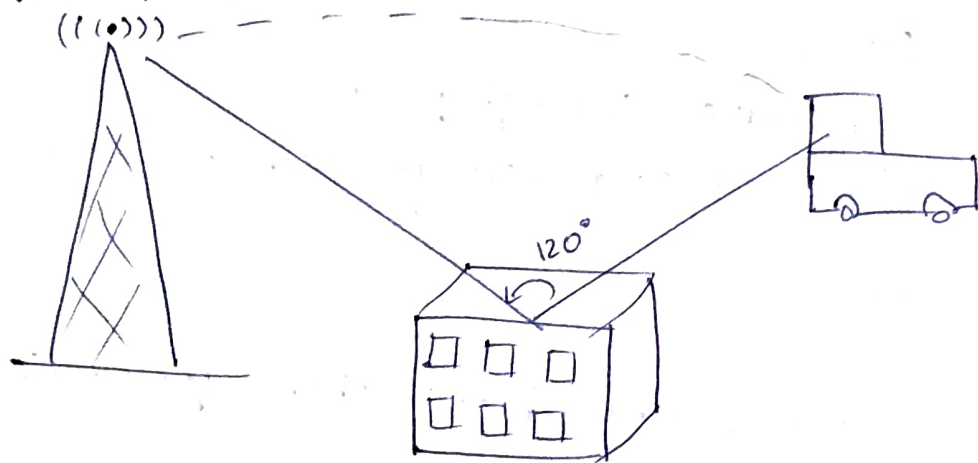
Basic Propagation Mechanisms



Diffraction happens when the network falls on sharp edges
 scattering happens when the Network falls on irregular surface.

◆ Reflection phase shift = 180°

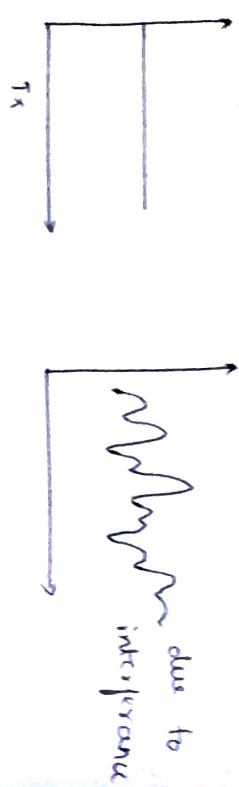
Effect of Reflection.



A wireless communication transmitter transmits a signal at 900MHz. A receiver is located at a distance 1km away from the transmitter. The receiver receives two signals one directly as LOS another indirectly via reflection from the building whose height is more than 10m

- variation in amplitude, phase and delay
- results because of multipath propagation
- Rapid fluctuation of the received signal strength

variation of the transmitted and received power



Note: If two signals are added - constructive interference

- If two signals is separated - destructive interference
- constructs & destructs due to phase

In phase = 0, Out of phase $\neq 0$.

- Mobility - when receiver changes the location. This effect is called as doppler effect.

Rayleigh fading } statistical models of

Rician fading } fading.

- Occurs when there are 'n' numbers of multipath
- No single dominant LOS
- Indoor
- Occur when there are (n-1) number of multipath

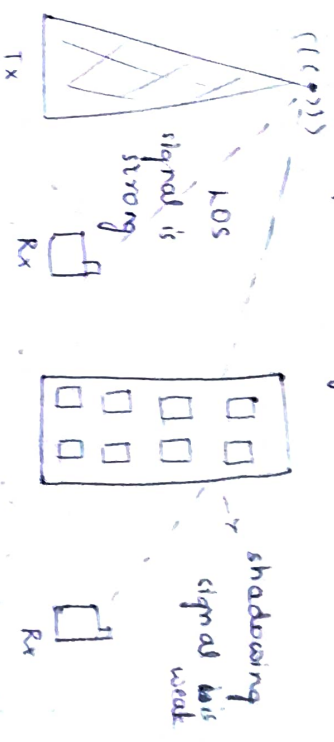
- single LOS
- Outdoor.

- small scale & large scale fading
- (variation of received signal w.r.t time or position)
- small scale fading - fluctuations of received signal w.r.t small scale change in time or position

→ Because of pathloss (reduction in received signal) $\text{pathloss} \propto \frac{1}{d^2}$

large scale fading - fluctuation of received signal w.r.t time or position in large scale

- Because of shadowing



Flat fading & frequency selective fading.

- flat fading occurs when the signal BW is less than the channel BW / coherence BW

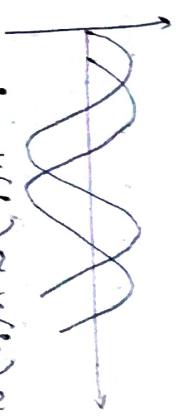
$B_s < B_c$

$B_s = 1 \text{ MHz} - \frac{1 \text{ MHz}}{99.5 \text{ MHz to } 100.5 \text{ MHz}}$

$B_c = 10 \text{ MHz}$

$f_1 = 99.5 \text{ MHz}$

$f_2 = 100.5 \text{ MHz}$



$f_1 = H(f_1) \approx H(f_2) \approx H(f_3)$

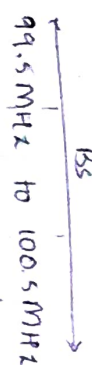
$f_0 = 100.5 \text{ MHz}$

∴ fading is uniform. Therefore it is flat fading.

Frequency selective fading: $B_c < B_s$

$$B_c = 0.2 \text{ MHz}$$

$$B_s = 1 \text{ MHz}$$



slow fading & fast fading.

Rapid variation
chadowing

→ slow variation.

A mobile subscriber travels at speed of 60 km/h compute the time between the fades if the mobile operates at two cases

a) 900 MHz

b) 1900 MHz

time between the fades is given by.

$$T_f = (\lambda c/2) / v_m = \lambda c / 2 v_m = c / (2 f v_m)$$

$v_m \rightarrow$ mobility of user

$$a) T_f = \frac{3 \times 10^8}{2 \times 900 \times 10^6 \times 60 \times 10^3}$$

$$T_f = 10 \text{ ms}$$

$$b) T_f = \frac{3 \times 10^8}{2 \times 1900 \times 10^6 \times 60 \times 10^3}$$

$$T_f = 4.7 \text{ ms}$$

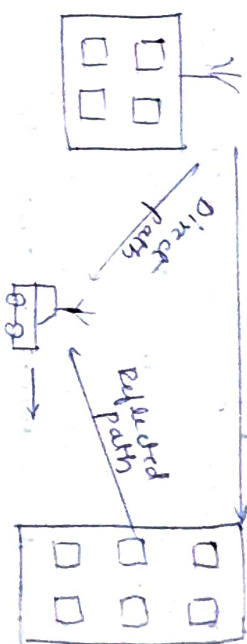
slow fading occurs when operates at 900 MHz and fast fading occurs when operates at 1900 MHz. It is observed that the rapidity of the fading increases with the frequency of transmission at the same speed of the mobile vehicle.

2) speed = 90 km/h

$$T_f = \frac{3 \times 10^8}{2 \times 900 \times 10^6 \times 90 \times 10^3}$$

$$60 \times 60$$

Rapidity not only increases with frequency. Rapidity also increases with mobility.



If the vehicle moves as the mobile moves forward at the distance $\lambda/4$ the direct path is increased and the reflected path is reduced by the same amount. so that resulting is a phase shift of 180° .

$$\frac{\lambda c}{4} + \frac{\lambda c}{4} = \frac{\lambda c}{2}$$

$$360/2 = 180^\circ$$

Delay spread:

$$71.09/26$$

consider when a signal is transmitted the receiver receives the copies of multipath components. The first path arrives at 10 μ s and the second path copy arrives at 18 μ s and the third copy from the third path arrives at 15 μ s with this we say that the delay spread of the transmitted signal

$$T_d = 5 \mu\text{s}$$

$\therefore$ Delay spread is the difference between earliest arrival and the latest arrival of the transmitted signal.

Propagation delay (T_p) - time taken to transmit 1 symbol
 Symbol duration (T_s) - time taken to transmit 1 symbol
 T_s < T_p → no delay, no interference
 T_s > T_p → ISI occurs (Inter-symbol Interference)

If the multipath delay path spread is greater than symbol duration then the received waveform spread out into neighbouring symbol and produces Inter-symbol Interference. This ISI results in irreducible error that are caused in the detected signal. So it affects detection at the receiver.

For a low bit error rate performance of a digital transmission the transmission data rate

$$T_s < \frac{1}{2T_d} \quad \left| \quad 10^{-3} \text{ s} \times 10^{-3} \times \frac{1}{10^4} = 10^{-7} \text{ s} \rightarrow \text{low bit error}$$

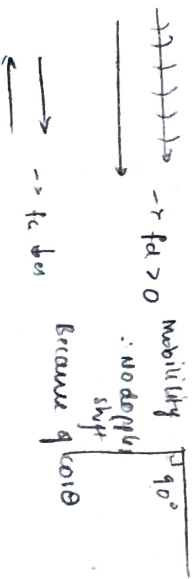
Typical value of delay spread is $\approx 3 \mu\text{s}$

Type of environment	Delay spread (s)
Inside the building	< 0.1
Open area	< 0.2
Sub urban area	< 0.5
Urban area	3.

path diff
 $d = 3 \times 10^8 \times 3 \times 10^{-4}$
 $= 900 \text{ m}$

Doppler shift:

$f_c = 900 \text{ MHz}$



- Consider a base station operating at 900 MHz carrier frequency for a mobile moving at a speed of 70 km/h. Determine the received carrier frequency if the mobile is moving
- directly away from the base station transmitter
 - directly approaching towards
 - In a direction which is 60° with a direction of arrival of transmitted signal.
 - In a direction of 45° to the direction of arrival of transmitted signal

Given: $f_c = 900 \text{ MHz}$

$$v_m = 70 \text{ km/h} = \frac{70 \times 10^3}{60 \times 60} = 20 \text{ m/s}^{-1}$$

a)



$$f_d = \left(\frac{f_c}{c} \right) v_m \cos \theta$$

$$f_d = \left(\frac{900 \times 10^6}{3 \times 10^8} \right) 20 = -60 \text{ Hz}$$

The received carrier frequency f_r at the

$$\text{mobile } f_r = f_c + f_d$$

$$= 900 \times 10^6 - 60$$

$$= 899.99994 \text{ MHz}$$

b)

$$f_r = f_c + f_d$$

$$= 900 \text{ MHz}$$

b) $f_d = \left(\frac{f_c}{c} \right) v_m \cos \theta$

$\theta = 0^\circ$

$f_d = 60 \text{ Hz}$

$f_c = 900.00006 \text{ MHz}$

c) $f_d = \frac{900 \times 10^6}{3 \times 10^8} \times 20 \times \frac{1}{2}$

$= 30 \text{ Hz}$

$f_r = 900 \times 10^6 + 30$

$= 900.00003 \text{ MHz}$

d) Since it is moving ^(mobile) direction far to the direction of arrival to the transmission signal it is making angle 90° then the cosine component turns 0. Then there is no doppler shift observed therefore the received signal frequency is same as the carrier signal frequency = 900 MHz .
solve problems on coherence time & BW

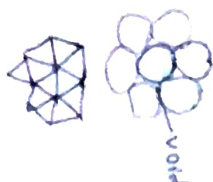
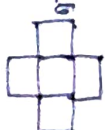
Principle of cellular communication:

cell - It is a geographical area in which coverage is provided.

→ void results in low coverage

→ triangle - no void but requires more no. of base station

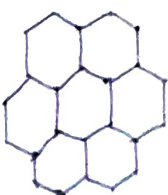
→ No void but not approximate of circle.



→ hexagon - approximate of circle

(radius = sides)

group of cell - cluster



cluster size,

$n = i^2 + j^2 + j^2$

where, $i, j \rightarrow$ non negative integers

$n = 3, (i=1, j=1)$

$i=1, j=2$

$n = 7$

$i=1, j=3, n = 13$

$i=2, j=2$

$n = 12$

$i=3, j=3, n = 27$

$i=3, j=1, n = 13$

$i=4, j=1, n = 21$

cluster of a cell gives more coverage area.

→ this main goal is to improve the spectrum efficiency.

cell structure & cluster:

Significance of cellular topology

consider a single high power transmitter than can support 40 voice channel over an area of

140 km^2 with the available spectrum. If this area is equally divided into 7 smaller areas

(cells) each supported by lower power transmitters so that each cell supports 30% of the channels

then determine.

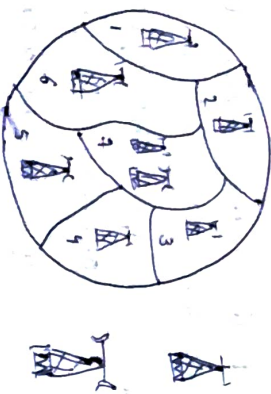
- coverage area of each cell
- Total number of voice channels available in a cellular system
- comment on the result spectrum.

Frequency reuse and spectrum efficiency 16/1/26
 considers a single high power freq that can support 100 voice channels which is divided as shown in fig. The available spectrum of 100 voice channels are (1-4) divided into 4 groups of 25 channels each then (1,4), (2,4), (3,5) and 6 are assigned distinct channel group with this shows that total number of channel that can be supported is enhanced to 175 to cover the same area.

Total number of voice channel $N = 4$ groups with each group 25 channels

- group 1 - (1, 4)
- group 2 - (2, 4)
- group 3 - (3, 5)
- group 4 - 6

$$K = 7$$



cell 1 and 7 are far enough and apart so that their signals do not cause unacceptable interference they cause acceptable interference. This called as frequency reuse, therefore they are co-channel.
 Total capacity = total no. of $N \times$ No. of channel per cell.

$$= 7 \times 25$$

$$\text{Percentage increased} = 75\%$$

$$\left| \frac{75}{100} \times 100 \right|$$

A mobile communication system is allocated RF spectrum of 25MHz and uses RF channel BW of 25KHz so that a total number of 1000 voice channels can be supported in a system.

a) If the service area is divided into 20 cells with the frequency reuse factor of 4, compute the system capacity.

b) The cell size is reduced to the extent that the service area is now covered with 100 cells. Compute the system capacity while keeping the frequency reuse factor of 4.

c) Consider the cell size is further reduced so that the same service area is now covered with 700 cells with the frequency reuse factor of 7. Compute the system capacity.

given Total available RF spectrum/BW = 25MHz

RF BW of each RF channel = 25KHz

$$N = \frac{\text{Total BW}}{\text{each channel BW}} = \frac{25M}{25K} = 1000 \text{ channels}$$

a) $K = 4$ (reuse factor/cluster size)

This means 4 cells together form a cluster and the 1000 channels are distributed among these 4 cells.

Each cell can accommodate 250 channels.

a) 20 cells, $K = 4$, 5 clusters.

$$\frac{20}{4} = 5 \text{ clusters.}$$

$$\text{Total capacity} = 1000 \times 5 = 5000$$

b) 100 cells $K = 4$

$$\frac{100}{4} = 25 \text{ clusters}$$

$$\text{Total capacity} = 25000$$

c) 700 cells $K = 7$

$$\frac{700}{7} = 100 \text{ clusters}$$

$$\text{Total capacity} = 100000$$

• No of channel per cell = $N = 7 \times K$ — (1)

$$7 = N/K$$

N → available channel

K → no of cells in one cluster

7 → no of channel per cell

$N = 100$ & cluster size $K = 7, 4, 3$. Determine various values for 7 .

$$7 = 100/4 = 14.25 \quad (\text{channel should be in whole no})$$

$$7 = 100/7 = 14.285$$

$$7 = 100/3 = 33.333$$

we this we can say that as the cluster size increases the channel capacity decrease.

Total system capacity $C = M \times N$ — (2)

C → total system capacity in channel

M → no of clusters in the entire service area

N → no of channels per cluster

Capacity in terms of 7 and K
 $C = M \times 7 \times K$

cellular system capacity. consider that a geographical service area of a cellular system is 4200 km^2 . A total of 1001 radio channels are available for handling traffic. Suppose the area of a cell is 12 km^2 .

a) How many times would the cluster of size 7 have to be replicated in order to cover the entire service area? Calculate the no. of channels per cell and the system capacity.

b) If the cluster size is decreased from 7 to 4 then does it result into increase in system capacity? Comment on the result obtained.

Area of total system $A_{\text{system}} = 4200 \text{ km}^2$
Total no. of available channel = 1001

$$A_{\text{cell}} = 12 \text{ km}^2$$

$$K = 7$$

$$\text{Area of cluster } A_{\text{cluster}} = K \times A_{\text{cell}}$$

$$= 7 \times 12 = 84 \text{ km}^2$$

$$\text{no. of clusters } M = \frac{A_{\text{system}}}{A_{\text{cluster}}} = \frac{4200}{84}$$

$$= 50 \text{ clusters}$$

$$J = \frac{N}{K} = \frac{1001}{7} = 143 \text{ channel/cell}$$

$$C = M \times N = 50 \times 1001 = 50050$$

b)

$$A_{\text{cluster}} = K \times A_{\text{cell}} = 4 \times 12 = 48 \text{ km}^2$$

$$M = \frac{A_{\text{system}}}{A_{\text{cluster}}} = \frac{4200}{48} = 87.5 \text{ clusters}$$

$$J = \frac{N}{K} = \frac{1001}{4} = 250$$

$$C = M \times N = 87 \times 1001 = 87087$$

Relationship b/w cluster size K and shift parameters i, j

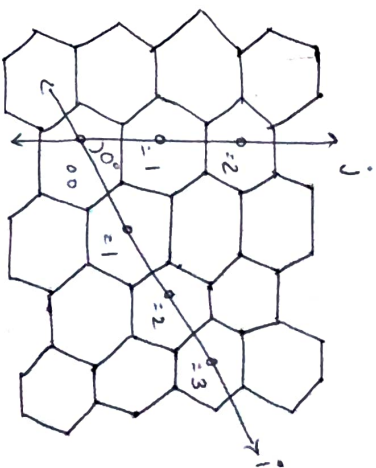


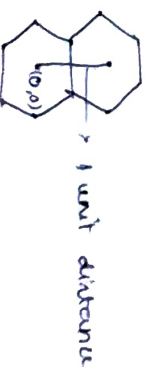
Fig. Shift parameters i and j in a hexagonal geometry.

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Method of locating co-channel cells.

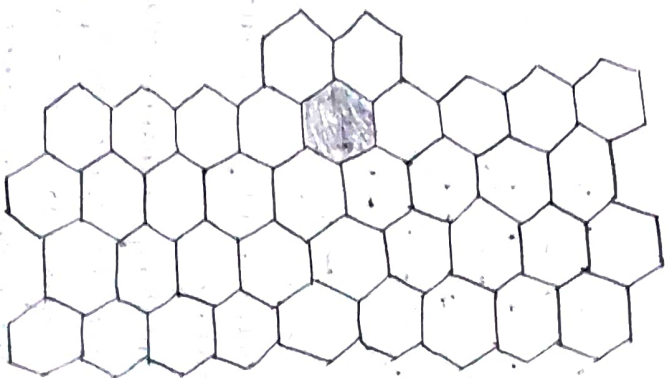
Cells which use the same set of frequency or channels are referred to as co-channel cells.

To locate the nearest co-channel cell mark the centre of the cell as $(0,0)$

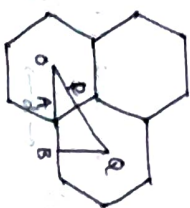


For which co-channel cells are required to be located by the unit distance as the distance of centre of two adjacent cells

Now follow the 2 step mentioned below.
 step 1: move i number of cells along any chain of hexagon.
 step 2: turn 60° ^{anticlockwise} clockwise and move j number of cells.
 the method of locating i=2 and j=3 is shown in fig



Prove that the distance between the centre of two adjacent regular hexagon is equal to $\sqrt{3}R$



↓ O-B distance b/w 2 co-located cells

Let R be the distance from the centre of a regular hexagon and any of its vertex.
 let d be the distance between the centres of two adjacent regular hexagons
 To show that $d = \sqrt{3}R$,
 from the geometry of the fig.

$$OA = R$$

$$AB = R/2$$

$$\text{then } OB = OA + AB$$

$$= R + R/2$$

$$= 3R/2$$

Then consider the right angle ΔOAP

$$\text{In } \Delta OAP, OP = OA \sin 60^\circ$$

WKT, from trigonometry the distance between the centre of two adjacent hexagonal cells.

$$OQ = OP + PQ$$

$$\therefore d = OA \sin 60^\circ +$$

$$= \left(\frac{\sqrt{3}}{2}\right)R + \left(\frac{\sqrt{3}}{2}\right)R$$

$$d = \sqrt{3}R$$

Frequency reuse ratio $q = D/E$

$q \rightarrow$ reuse ratio

$D \rightarrow$ distance between two nearest co-channel cells

$R \rightarrow$ radius of cell

Determine the distance from the nearest co-channel cell from the cell having a radius of 0.64 km and a co-channel reuse factor of 12.

$$q = D/R \Rightarrow D = Rq$$
$$= 0.64 \times 12$$

$$D = 7.68 \text{ km}$$

Determine the frequency reuse ratio for a cell of radius 0.8 km separated from the nearest co-channel cell by a distance of 6.4 km

$$q = D/R$$

$$q = 6.4 / 0.8$$

$$q = 8$$

The frequency reuse ratio q is related to the cluster size as $q = \sqrt{3K}$. Determine various values of q for various values of K , 3, 4, 7, 9, 12, 13

$$K = 3, \quad q = \sqrt{3 \times 3} = \sqrt{9} = 3$$

$$K = 4, \quad q = \sqrt{3 \times 4} = \sqrt{12} = 3.46$$

$$K = 7, \quad q = \sqrt{3 \times 7} = \sqrt{21} = 4.58$$

$$K = 9, \quad q = \sqrt{3 \times 9} = \sqrt{27} = 5.19$$

$$K = 12, \quad q = \sqrt{3 \times 12} = \sqrt{36} = 6$$

$$K = 13, \quad q = \sqrt{3 \times 13} = 6.245$$

Determine the no. of cells in the cluster for the following values of i, j , (2,3), (4,1), (3,3), (5,0)

$$N = i^2 + ij + j^2$$

$$i = 2, j = 3, \quad N = 2^2 + 2 \times 3 + 3^2$$

$$N = 19$$

$$i = 4, j = 1, \quad N = 4^2 + 4 \times 1 + 1^2 = 21$$

$$i = 3, j = 3, \quad N = 3^2 + 3 \times 3 + 3^2 = 27$$

$$i = 5, j = 0, \quad N = 5^2 + 0 \times 0 + 0^2 = 25$$